

Effective line redistribution in kilonova ejecta: transport-relevant structure is frequency-local

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Abstract

Lanthanide line fluorescence in kilonova ejecta couples of order 10^6 transitions, and radiative-transfer calculations therefore replace it with a closure. On identical resolved Sobolev opacity, with an energy-conserving downward macroatom as the reference, we show that no single thermalisation parameter ϵ reproduces the emergent photometry, while a group-to-group redistribution operator with a handful of contiguous frequency groups does. Comparing local frequency coarsening with a global non-negative factorisation at a matched number of archetypal exit distributions, the local representation reaches the 0.10 mag criterion with no more archetypes than the global one, and on Nd II with far fewer; at 32 archetypes the global operator reproduces the microscopic redistribution better while reproducing the light worse. The structure that matters for transport is frequency-local. It is not, however, universal: an operator built at one thermodynamic state does not remain predictive when the gas temperature, density or the physical trajectory changes, whereas an operator rebuilt at the new state does, and interpolating the whole operator between two neighbouring states recovers most of the difference — but not along a coupled trajectory through state space, and not at a fixed resolution for Ce II. Per-species operators do not compose into a mixture without refitting, yet the full 13-ion mixture itself is described by 4 groups where the best scalar closure misses by 1.69 mag. Detailed lanthanide fluorescence thus has a much lower observable complexity than its microscopic transition network: the reduced description is frequency-local and remains compact across physical states and realistic mixtures, but its parameters, and sometimes its required resolution, depend on the thermodynamic and radiation-field state.

1 Introduction

The optical and infrared light of a kilonova is shaped by the line opacity of the lanthanides synthesised in the ejecta, whose open f -shells carry of order 10^6 transitions per ion at the relevant temperatures [Barnes and Kasen, 2013, Tanaka et al., 2020]. A photon absorbed in one of those lines is re-emitted, after a cascade through the ion’s levels, in another; this fluorescence redistributes energy across the spectrum and is the dominant non-local process in the transport. No calculation follows every transition. Every kilonova radiative-transfer code replaces the fluorescence by a closure: a single thermalisation parameter ϵ that sends a fraction of the absorbed energy to a thermal pool and scatters the rest coherently [Fontes et al., 2020], an expansion-opacity treatment that groups the lines and prescribes the re-emission within a group, or a macroatom that walks the cascade explicitly [Lucy, 2002, 2003]. In a companion methods paper we showed, on published benchmark ejecta states and with an energy-conserving treatment of the fluorescence, that the coarse line-opacity closures leave errors of order a magnitude in the broad-band photometry and that the errors change shape when the fluorescence prescription changes (Paper IV). The ques-

tion that leaves open is whether the fluorescence itself — not the opacity — admits a compact description at all, and if so what it depends on.

We ask that question in the narrowest form we can control. The opacity is held fixed at the resolved Sobolev value of every line; the reference is an energy-conserving downward macroatom; and the candidate closure is a redistribution operator R_{ij} that maps the frequency group of an absorbed packet to the frequency group and exit line of the re-emitted one. The operator is built from the reference’s own Monte Carlo events on one set of random seeds and scored on another, so that no operator is ever compared with the events it was trained on. Three questions follow one another. Does such an operator exist, and is it better than the best scalar closure? Why does it work — is the structure it exploits locality in frequency, or a globally low rank of the microscopic redistribution law? And how far does it generalise: across thermodynamic state, across the radiation field, and across composition?

Each question was posed as a preregistered gate with its thresholds, its gray conditions and its readings fixed before the run, and the record of every run, every reading and every amendment made during execution is public (Section 2). The answers are, in order: a frequency-local operator with a handful of groups reproduces the reference where no scalar closure comes within a magnitude (Section 3); the compression is frequency locality, and a global representation that reproduces the microscopic events better reproduces the emergent light worse (Section 4); and the operator is compact everywhere but universal nowhere — its parameters, and for one ion its required resolution, depend on the state, in a way that a table over a few coordinates captures along single coordinates but not along a coupled trajectory, while the realistic mixture remains as compact as any single ion (Section 5). The result is a hierarchy: millions of transitions, to a fine distribution of redistribution events, to a small state-dependent local operator, to the observables — with most of the information discarded between the middle two steps and no harm done to the last.

2 Method

State and atom. Every calculation is a single-zone, time-independent transport through one shell of a published kilonova ejecta model at two days — the photospheric shell of the lanthanide-rich secular component used as benchmark P1 in Paper IV — with the shell’s own density, gas temperature and expansion time, and a black-body core at the gas temperature illuminating it. The atom is one ion at its number density in that shell, its levels and lines from the GSI calibrated line list [Flörs et al., 2026], level populations in local thermodynamic equilibrium, and every line with Sobolev optical depth above 10^{-3} carrying opacity. La II, Ce II and Nd II are the three ions; the first is thin enough at its partial density that every closure agrees, and is carried as a control. The generality test of Section 5 moves one coordinate of this state at a time and, for the mixtures, replaces the single ion by the three ions together and by the state’s full 13-ion composition.

Transport and reference. Packets carry indivisible energy and are transported with world-line relativity and Sobolev line interaction (Paper IV). The reference fluorescence is a downward macroatom: on absorption the packet activates the line’s upper level and walks the radiative cascade with branching probabilities $A\beta$, the escape probability β of each line included, until it exits in a line whose photon escapes; energy is conserved at every step by construction, and the identity between injected and accounted energy is verified to 10^{-10} on every run. A radiation-field-driven macroatom with internal upward transitions under an imposed diluted Planck field was run once as a robustness check (Section 3), and a scalar closure ϵ — thermal re-emission from the energy-weighted line emissivity with probability ϵ , coherent scattering otherwise — was scanned over a

grid of values to find the best one for each ion.

The operator. The redistribution operator is built from the reference’s events, each event the absorbed comoving frequency, the re-emitted rest frequency and the packet’s energy before and after. On N_g log-spaced frequency groups the energy transition matrix R_{ij} is the fraction of energy absorbed in group i that exits in group j ; within each output group the exit line is drawn from a discrete table of the distinct exit lines the events populated, weighted by their energy. Transport with the operator replaces the macroatom walk by one draw from R_{ij} and one from the table. Operators are built from events on three build seeds and scored on three disjoint evaluation seeds. The global family of Section 4 replaces the transition matrix by a rank- k non-negative factorisation of the fine-grouped matrix, iterated to convergence, on the same exit tables; the interpolated operators of Section 5 combine two endpoint operators linearly in the logarithm of the moved coordinate, transition rows, deposition channel and exit tables alike, on the union of the two endpoints’ exit lines.

Metrics and gates. Every closure is scored against the reference on the same state and seeds by the maximum and mean absolute error in the broad-band magnitudes over the bands that carry at least one per cent of the luminosity with acceptable Monte Carlo precision, by the maximum absolute error in the colours between those bands, by an integrated spectral distance, and by an energy-weighted total-variation distance between the operator and an independent fine-grouped matrix from the evaluation events. The criterion used throughout is a maximum band error of 0.10 mag and a maximum colour error of 0.10 mag. Each of the three questions was preregistered as a gate — the state, the grids, the metrics, the thresholds, the gray conditions that invalidate a reading, and the readings themselves fixed and committed before the run — and read gray-first. Three amendments were made during the runs: two implementation defects in the factorisation were found on the control ion and fixed before any decisive number was read; a validity control fired and its preregistered remedy was applied; and a sampler overflow found on the mixture run was fixed, its reachability in every earlier record established by inspection, and the affected calculation repeated. Each is recorded, with what had been seen when it was made, in the archived preregistration.

3 A scalar closure is insufficient; a small operator is not

The optimally tuned scalar misses the reference by 1.40 mag on Ce II and 1.06 mag on Nd II, while the group operator meets the 0.10 mag criterion at $N_g = 16$ and $N_g = 2$ respectively (Fig. 1a, Table 1). La II at its partial density in this state interacts 0.7 times per packet and does not discriminate the closures; it is carried as a control throughout.

The result is not a property of one cascade prescription. Replacing the reference by a radiation-field-driven macroatom with internal upward transitions moves the emergent photometry by 2.09 mag on Ce II and 0.89 mag on Nd II; an operator rebuilt from that reference’s own events follows it to 0.026 and 0.046 mag, while the operator trained on the original reference stays 1.96 mag away, reproducing the physics it was built from (Fig. 1b).

4 Why it works: locality, not low rank

Two families of operator were built on identical exit tables and identical opacity, differing only in the group-to-group matrix: local coarsening into N_g contiguous frequency blocks, and a global non-negative factorisation into k freely mixed archetypes. At a matched count the global family is

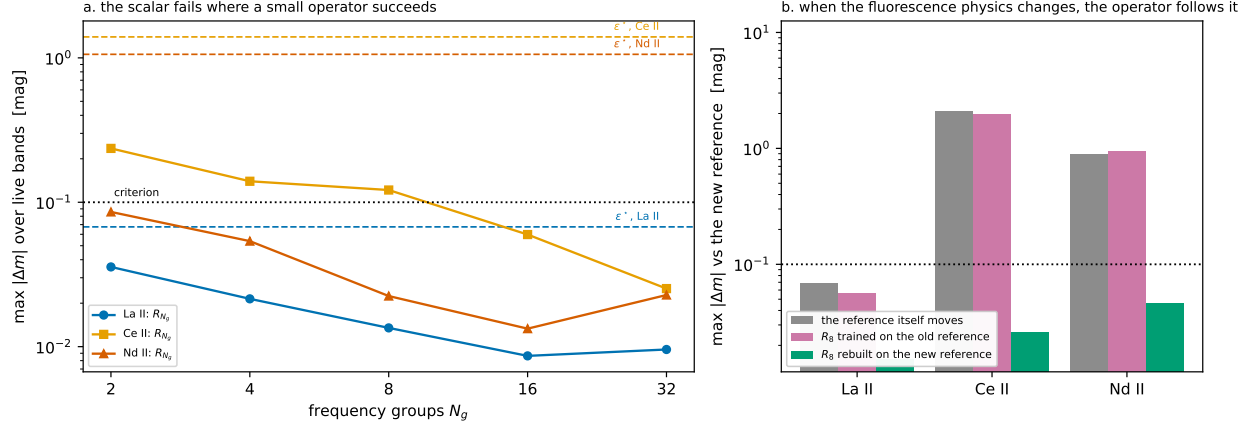


Figure 1: **A scalar closure fails where a small operator succeeds, and the operator follows the fluorescence physics.** (a) Maximum photometric error against the energy-conserving downward macroatom, over the live bands, for the group operator at $N_g = 2 \dots 32$ (solid) and for the optimally tuned scalar ϵ^* (dashed). (b) When the reference is replaced by a radiation-field-driven macroatom the reference itself moves (grey); an operator rebuilt from its events follows it (green), while the operator trained on the previous reference does not (pink).

Table 1: The gates, per ion.

ion	N_g^*	ϵ^*	error at ϵ^* [mag]	K_{local}^*	K_{global}^*	L^*	R_8 vs the new reference [mag]
La II	2	0.20	0.07	2	2	0.504	0.015
Ce II	16	0.00	1.40	16	32	0.358	0.026
Nd II	2	0.00	1.06	2	undefined	0.113	0.046

far the more expressive, carrying 8,192 free numbers against 1,024 at 32 archetypes. It nonetheless needs no fewer archetypes to reach the criterion: $K_{\text{local}}^* = 16$ against $K_{\text{global}}^* = 32$ on Ce II, and $K_{\text{local}}^* = 2$ on Nd II against a global family that is undefined at every rank tried.

The contrast runs the other way on the microscopic distribution (Table 2, Fig. 2). At 32 archetypes the global operator’s event-level total-variation loss is 0.075 against the local operator’s 0.132 on Nd II — it reproduces the microscopic redistribution 1.8 times better — while its photometric error is 0.162 mag against 0.008 mag, 20.8 times worse. The degrees of freedom that minimise a generic distance between microscopic redistribution distributions are therefore not the degrees of freedom that carry the radiation field.

The compressibility is two-level. Writing $P(\nu_{\text{out}} | \nu_{\text{in}}) = P(j | i) P(\nu_{\text{out}} | i, j)$, the first factor is extremely coarse, while the second still needs a sparse but appreciable set of real exit lines: truncating the exit tables by retained energy, the emergent light needs 0.358 of the distinct exit lines on Ce II and 0.113 on Nd II at the same thresholds. The transport-relevant organisation of fluorescence is low-dimensional and frequency-local, even though the microscopic exit spectrum is not equally compressible.

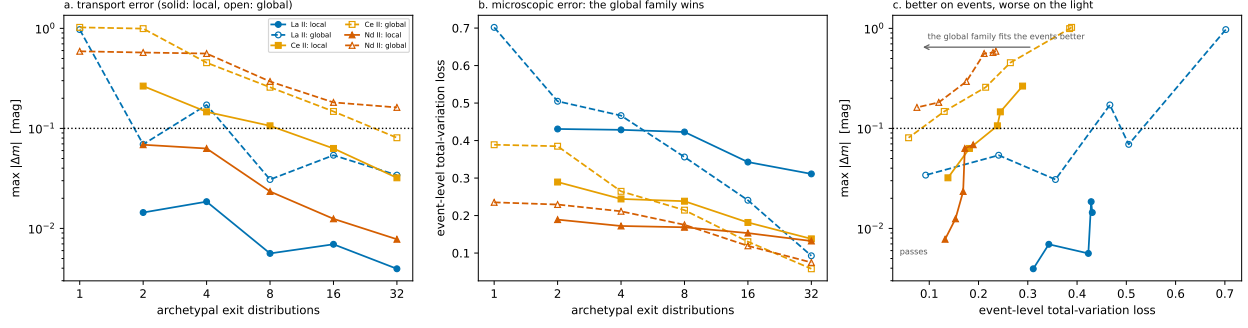


Figure 2: **Better on the events, worse on the light.** (a) Photometric error and (b) event-level total-variation loss against the number of archetypal exit distributions, for local frequency coarsening (solid) and the global factorisation (open). (c) The two against each other: the global points move left, fitting the microscopic redistribution better, while staying high, reproducing the emergent light worse.

Table 2: Microscopic against observable error at 32 archetypes.

ion	m_{event} global	m_{event} local	$\max \Delta m $ global [mag]	$\max \Delta m $ local [mag]
La II	0.093	0.311	0.034	0.004
Ce II	0.058	0.138	0.081	0.032
Nd II	0.075	0.132	0.162	0.008

5 Generality: a compact operator that depends on the state

The first three sections concern one thermodynamic state. The last question is whether the operator built there is a property of the ion or of the state it was built in. We moved one coordinate at a time away from that state along 4 axes — the gas temperature that sets the level populations, the number density that sets the optical-depth scale, the temperature of the illuminating source that sets which input groups the light actually uses, and the physical trajectory of the ejecta at one, three and five days, along which every coordinate moves together — 14 states per ion in all, on one frequency support per ion frozen over the whole domain. At every state two operators were scored against that state’s own reference (Fig. 3, Table 3): the operator built at the original state and transported unchanged, which tests *transfer*, and an operator rebuilt from the new state’s own events, which tests whether the compact description still *exists* there. The two tests were kept apart by design, because their failures mean different things. Both use 16 groups for every ion, one controlled representation size across the comparison.

Transfer fails; compression does not. The transported operator is not predictive. For Ce II it fails the criterion on every axis; for Nd II and the La II control it survives only along the source-spectrum axis, the one coordinate that changes which rows of the operator are illuminated without changing the atom’s response. Anywhere the populations or the optical depths move, an operator trained elsewhere is simply the wrong operator: 0.150 mag on Nd II at the interior temperature point, midway in $\log T_{\text{gas}}$ between two states at which the operator was rebuilt, and larger at the ends of every axis. That kills the strongest reading of Sections 3–4: there is no universal R_{ij} . But the operator rebuilt at the new state passes at every one of the 12 readable Nd II states and every one of the 14 La II states, with at most 4 groups needed anywhere on Nd II. The failure of transfer

is not a failure of compression. What varies with the state is the content of a small operator, not its size.

Interpolation: $R = R(\theta)$. If the operator’s parameters vary smoothly with the state, a table over a few coordinates should replace it. At the interior grid point of each axis we built an operator from the two bracketing states alone — never from the interior state’s own events — by interpolating the whole effective operator linearly in the logarithm of the moved coordinate: the transition rows, the deposition channel, and the conditional exit-frequency weights on the union of the two endpoints’ exit lines. That interpolant passes the criterion at every interior point on the temperature, density and source-spectrum axes for every ion, reducing Nd II’s transfer error at the interior temperature from 0.150 to 0.032 mag and Ce II’s at the interior density from 0.223 to 0.047 mag. The physically useful object is therefore a state-dependent family $R(\theta)$ over a small set of coordinates, and along a single coordinate that family is smooth enough to tabulate. The conditional exit spectrum is part of what must be tabulated: interpolating the transition matrix alone while keeping one endpoint’s exit tables leaves Ce II at 0.130 mag on the density axis, above the criterion, where the whole-operator interpolant passes at 0.047.

The coupled trajectory is the boundary of that statement. The interpolation test above moves one coordinate at a time. Along the physical trajectory, where temperature, density and the source all change together, Nd II at three days is the one point in the domain where interpolation fails: the transported operator misses by 0.181 mag, the whole-operator interpolant by 0.117 mag, and the matrix-only interpolant fails as well, on colour rather than on band — while the operator rebuilt at that state passes at 2 groups with 0.013 mag. This is not a failure of compactness; it is a failure of separable interpolation. The operator manifold cannot necessarily be reconstructed by treating the state coordinates independently, and we do not claim that interpolation works in general: it was tested primarily one axis at a time, and the one coupled path tested breaks it.

The required resolution is itself state-dependent. Ce II adds a separate qualification. Its rebuilt 16-group operator fails the criterion at 7 of its 14 states — at every one of them on colour, by between 0.007 and 0.062 mag over the threshold, and at 1 of them on band as well — and at every one of them a 32-group operator passes. Section 3 found Ce II’s $N_g^* = 16$ exactly at the edge of the criterion at the original state; across the domain that marginality becomes a resolution requirement that moves with the state, $N_g^*(\theta)$. Compactness survives, but a fixed representation size does not, and a practical table for Ce II needs either adaptive resolution or the conservative 32-group representation throughout. La II, whose interactions per packet are too few to discriminate any closure, passes everywhere and decides nothing.

Composition. Two mixture tests close the gate. First, at the original state the three single-ion operators were combined by the opacity-share composition rule of the earlier photon-packet work, with no fit to the blend, and scored against the three-ion blend’s own macroatom: the composed operator does not reach the criterion at any valid tested resolution (0.142 mag at its best, 16 groups; the finest-resolution run was excluded by the preregistered validity check), while an operator trained on the blend’s own events passes at 4 groups (0.083 mag). Per-species operators are not trivially composable. Second, and decisively, on the full 13-ion composition of the published benchmark state the operator built from the blend’s events passes at 4 groups (0.088 mag on band, 0.073 mag on colour), while the best scalar closure — again the coherent limit, $\epsilon^* = 0.00$ — misses by 1.69 mag, the widest scalar failure in this work. Read together, the two results are not in tension:

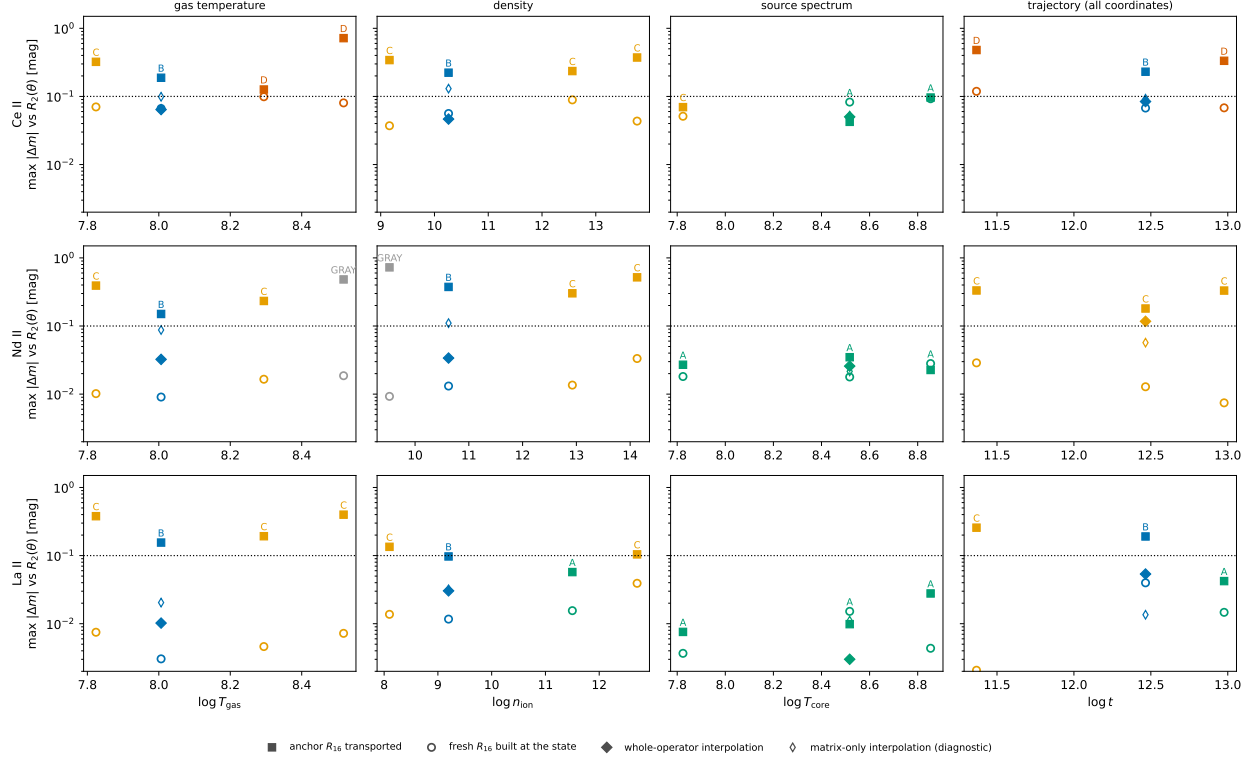


Figure 3: **Transfer is not compression.** Maximum photometric error against the state’s own reference for the operator built at the original state and transported unchanged (filled squares), the operator rebuilt at the state (open circles), and at each axis’ interior point the whole-operator interpolant built from the two bracketing states (filled diamonds; the matrix-only interpolant as open diamonds). Rows: Ce II, Nd II, the La II control; columns: the gas temperature, the density, the source spectrum, and the physical trajectory along which every coordinate moves together. The dotted line is the band criterion; the letter is the preregistered reading of each state (A: the transported operator passes; B: only the interpolant passes; C: only a fresh fit passes; D: none passes at 16 groups). The criterion also bounds the colour error, so a circle below the line can still read D. Grey: states excluded by the preregistered validity checks.

the combined physical system is itself highly compressible even though it cannot be assembled from its parts without refitting. It is this result that keeps the state dependence above from reducing the paper to a negative finding about transfer. The physically relevant mixture still admits a small effective description.

6 Discussion and conclusions

The three gates together establish a hierarchy. Of the order 10^6 transitions of a lanthanide ion, the transport sees a fine distribution of redistribution events; of that distribution, the emergent light sees a small operator; and that operator is local in frequency. Most of the information in the event distribution is discarded on the way to the light without harm — the global factorisation of Section 4 reproduces the events 1.8 times better than the local operator and the photometry 20.8 times worse — so fidelity to the microscopic redistribution is the wrong target for a closure. The

Table 3: Generality, per ion: the preregistered readings, the axes along which the transported operator stays predictive, how often the rebuilt 16-group operator fails and the largest resolution it needed, and the interior-point interpolation error (whole operator / transition matrix only) on the three single-coordinate axes, in mag.

ion	C1	C2	C3	transfer holds (axes)	fresh R_{16} fails (of states)	max K_{rec}^*	interp. T whole / matrix	inte whole
La II	GREEN	YELLOW	GREEN	J	0/ 14	2	0.010/ 0.020	0.030
Ce II	RED	YELLOW	YELLOW	none	7/ 14	32	0.064/ 0.099	0.047
Nd II	GREEN	RED	GREEN	J	0/ 12	4	0.032/ 0.087	0.034

right target is the frequency-local structure, and a handful of contiguous groups carries it.

What Section 5 adds is that the small operator is not one object but a family. Its parameters depend on the gas temperature, the density and the illumination, and the dependence is smooth enough along any one of those coordinates that an operator interpolated between two neighbouring states passes the same criterion as one built in place. That is the practically useful form of the result: not a universal R_{ij} , but a tabulated $R(\theta)$ over a few state coordinates, of the same size at every entry. Two boundaries of that statement are set by the data and should be read as results rather than caveats. Along the physical trajectory, where the coordinates move together, one-dimensional interpolation fails at the one interior point tested while a fresh fit succeeds — the family is compact but not separable, and a table indexed by independent coordinates is not guaranteed to reconstruct it. And for Ce II the resolution the criterion requires is itself state-dependent: 16 groups suffice at the original state and fail at half of the others, 32 groups pass everywhere. A closure of this kind must carry its resolution as a parameter or adopt the conservative one.

The mixture results settle how the family composes. It does not: operators built for single ions, combined by the opacity-share rule that served the earlier photon-packet work, do not reproduce the blend they are combined into, while an operator built on the blend does, at 4 groups. Yet the full 13-ion composition of the benchmark state — the mixture a transport calculation actually needs — is described by 4 groups where the best scalar closure misses by 1.69 mag. Per-species operators are not composable, but the combined system is compressible, and the effective description a code would use is the one built on the mixture it transports.

Three limitations bound the claims. The calculations are single-zone and time-independent, with the level populations in local thermodynamic equilibrium and the radiation field imposed rather than solved; the operator’s dependence on a self-consistent radiation field is measured only through the source-spectrum axis. The state domain is a grid of single coordinates around one benchmark state and one coupled path; the failure of interpolation along that path is one data point, and it does not yet indicate what representation would succeed there — a better choice of coordinates or a denser table are the first candidates, ahead of any learned surrogate. And the criterion is photometric: an operator that passes it reproduces the broad-band light, not the line-by-line spectrum.

The conclusion we draw is nevertheless a positive one for transport. Detailed lanthanide fluorescence has a much lower observable complexity than its microscopic transition network. The reduced description is frequency-local, survives a change of the fluorescence physics it is built from, and remains compact across physical states and realistic mixtures. What it does not do is transfer: its parameters, and sometimes its required resolution, depend on the thermodynamic and radiation-field state, and a transport code that adopts it must carry that dependence — as a table along the coordinates where the dependence is smooth, and as a refit where it is not.

Data availability

The transport code, the preregistrations of gates G1–G3, every run record and the script that regenerates every number in this paper are archived at github.com/zhuygln/revisit_sobolev (tags `paperB-g2-prereg`, `paperB-g2-freeze`, `paperB-g3-prereg` and `paperB-g3-freeze`).

References

- J. Barnes and D. Kasen. Effect of a high opacity on the light curves of radioactively powered transients from compact object mergers. *The Astrophysical Journal*, 775(1):18, 2013. doi: 10.1088/0004-637X/775/1/18.
- Andreas Flörs, Ricardo Ferreira da Silva, José Pires Marques, Jorge Miguel Sampaio, and Gabriel Martínez-Pinedo. Calibrated lanthanide atomic data for kilonova radiative transfer: Atomic structure and opacities. *Physical Review D*, 113(6):063041, 2026. doi: 10.1103/jxqw-7ynk.
- C. J. Fontes, C. L. Fryer, A. L. Hungerford, R. T. Wollaeger, and O. Korobkin. A line-binned treatment of opacities for the spectra and light curves from neutron star mergers. *Monthly Notices of the Royal Astronomical Society*, 493(3):4143–4171, 2020. doi: 10.1093/mnras/staa485.
- L. B. Lucy. Monte carlo transition probabilities. *Astronomy & Astrophysics*, 384:725–735, 2002. doi: 10.1051/0004-6361:20011756.
- L. B. Lucy. Monte carlo transition probabilities. II. *Astronomy & Astrophysics*, 403:261–275, 2003. doi: 10.1051/0004-6361:20030357.
- M. Tanaka, D. Kato, G. Gaigalas, and K. Kawaguchi. Systematic opacity calculations for kilonovae. *Monthly Notices of the Royal Astronomical Society*, 496(2):1369–1392, 2020. doi: 10.1093/mnras/staa1576.